

EXPERIMENT-2

Student Name: Hunar Katyal

UID: 24BDA70085

Branch: AIT-CSE (Data Science)

Section/Group: 24AIT_KRG-G2

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Subject Name: Computer Networks

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AIM: Study the basic network command and Network configuration commands like ping, variations of ip config, tracert, nslookup, netstat, arp, rarp, hostname, pathping and basic networking commands.

- ping
- ipconfig
- tracert
- nslookup
- netstat
- arp
- hostname
- pathping
- route
- net use

OBJECTIVES:

- To study the basic Windows networking commands used for network configuration and troubleshooting.
- To understand the working of the Ping command and its variations for testing network connectivity using ICMP.
- To learn the use of IPConfig commands for viewing and managing TCP/IP configuration.

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- To understand the Tracert command for tracing the route taken by packets from the source to the destination.
- To study the NSLookup command for DNS name resolution and reverse lookup.
- To examine active network connections, listening ports, and protocol statistics using the Netstat command.
- To understand the ARP command for viewing and managing the ARP cache (IP-to-MAC address mapping).
- To learn the use of the Hostname, PathPing, Route, and Net Use commands for identifying the system, analyzing network paths, viewing routing information, etc.

SOFTWARE REQUIREMENTS:

(a) Software:

- Operating System
- Terminal

(b) Hardware (Virtual):

- Desktop Computer or Laptop
- Network Interface Card (NIC)
- Internet/LAN Connection

THEORY: (i) Ping

Ping (Packet Internet Groper) is a network diagnostic utility used to verify whether a remote host or network device is reachable over an IP network. It works by sending **ICMP (Internet Control Message Protocol) Echo Request** packets to the destination host and waiting for **ICMP Echo Reply** packets. If the destination responds, the command displays the round-trip time (RTT), packet loss, and Time To Live (TTL) value. Ping is one of the most commonly used tools for testing network connectivity and troubleshooting communication problems.

Working of Ping

1. The source computer sends an ICMP Echo Request packet to the destination.
2. The destination receives the packet and sends back an ICMP Echo Reply.
3. The source measures the time taken for the packet to travel to the destination and back.

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4. If no reply is received within a specified time, the request is considered timed out.

Purpose of Ping

- To verify whether a host is reachable.
- To test network connectivity.
- To measure response time (latency).
- To detect packet loss.
- To troubleshoot network problems.

Common Variations

- ping <IP/hostname> – Tests connectivity.
- ping -t – Sends continuous Echo Requests until stopped.
- ping -n <count> – Sends a specified number of Echo Requests.
- ping -l <size> – Specifies the size of the data payload.
- ping -f – Sets the "Don't Fragment" flag.
- ping -a – Resolves an IP address into its hostname.

(ii) IPConfig

IPConfig (Internet Protocol Configuration) is a Windows command-line utility used to display and manage the TCP/IP configuration of network interfaces. It provides information such as the IP address, subnet mask, default gateway, DHCP status, and DNS server details. It is an essential command for verifying network configuration and troubleshooting IP-related issues.

Purpose

- Display network configuration.
- Verify IP address assignment.
- Check subnet mask and gateway.

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- Renew or release DHCP addresses.
 - Troubleshoot IP connectivity problems.
-

(iii) Tracert

Tracert (Trace Route) is a Windows networking command that traces the complete path taken by packets from the source computer to the destination. Unlike Ping, which only confirms whether a destination is reachable, Tracert identifies every intermediate router (hop) through which the packets travel. It uses ICMP Echo Request packets with increasing TTL (Time-To-Live) values.

Working

- Sends packets with TTL = 1.
- The first router decreases TTL to zero and returns an ICMP Time Exceeded message.
- TTL is gradually increased to discover each router along the path.
- The process continues until the destination is reached.

Purpose

- Display the routing path.
 - Identify slow routers.
 - Detect routing failures.
 - Troubleshoot Internet connectivity.
-

(iv) NSLookup

NSLookup (Name Server Lookup) is a command-line utility used to query the Domain Name System (DNS). DNS translates human-readable domain names into IP addresses. NSLookup helps verify DNS functionality and diagnose DNS-related issues.

Purpose

- Resolve domain names to IP addresses.

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- Perform reverse DNS lookup.
 - Verify DNS server operation.
 - Retrieve different DNS record types.
-

(v) Netstat

Netstat (Network Statistics) is a command-line utility that displays network-related information such as active connections, listening ports, routing tables, protocol statistics, and Ethernet statistics. It is widely used by network administrators to monitor network activity and identify suspicious or unnecessary connections.

Purpose

- View active TCP/UDP connections.
 - Display listening ports.
 - Show routing information.
 - Display protocol statistics.
 - Identify process IDs using network ports.
-

(vi) ARP

ARP (Address Resolution Protocol) is used to map an IPv4 address to its corresponding MAC address on a local network. Communication over Ethernet requires MAC addresses, while applications use IP addresses. ARP bridges this gap by maintaining an **ARP Cache**.

Working

1. The sender knows the destination IP.
2. It broadcasts an ARP Request.
3. The destination replies with its MAC address.
4. The sender stores the mapping in the ARP cache.

Purpose

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- View ARP cache.
 - Delete ARP entries.
 - Create static ARP entries.
 - Troubleshoot LAN communication.
-

(vii) Hostname

The **Hostname** command displays the name assigned to the current computer. Every computer connected to a network has a unique hostname for easy identification.

Purpose

- Identify the computer on the network.
 - Assist in network administration.
 - Verify the system name.
-

(viii) PathPing

PathPing combines the features of **Ping** and **Tracert**. It traces the complete path to the destination and then sends multiple ICMP Echo Requests to each router to calculate packet loss and network latency at every hop.

Purpose

- Detect packet loss.
- Identify slow routers.
- Analyze network latency.
- Troubleshoot network performance.

Working

1. Finds the route like Tracert.
2. Sends multiple ping packets to every hop.

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3. Calculates packet loss percentage.
 4. Displays latency and delay.
-

(ix) Route

The **Route** command is used to display and manage the routing table in Windows. A routing table contains information about the paths used to forward packets to different networks.

Purpose

- Display routing table.
 - Add static routes.
 - Delete routes.
 - Modify routing entries.
-

(x) Net Use

Net Use is a Windows networking command used to connect, disconnect, and manage shared network resources such as shared folders and printers. It allows users to map a network share to a drive letter for easier access.

Purpose

- Display shared network resources.
- Map network drives.
- Disconnect mapped drives.
- Manage network connections.

STEPS/IMPLEMENTATION:

Part I – Ping Command

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1. Open the Command Prompt (CMD) by pressing Windows + R, typing cmd, and pressing Enter.
2. Execute the Ping command to check network connectivity.
3. Observe the reply time, TTL value, and packet loss.
4. Execute different Ping variations such as -t, -n, -l, -f, and -a to understand their functions.

Part II – IPConfig Command

6. Execute the ipconfig command to view the current IP configuration.
7. Execute ipconfig /all to display detailed TCP/IP configuration.
8. Execute ipconfig /release to release the current DHCP-assigned IP address.
9. Execute ipconfig /renew to obtain a new IP address from the DHCP server.
10. Observe the IP address, subnet mask, default gateway, DNS server, and MAC address.

Part III – Tracert Command

11. Execute the tracert command to trace the route to a destination.
12. Observe the number of hops, router IP addresses, and response time for each hop.
13. Execute tracert -d to display only IP addresses.
14. Execute tracert -h to specify the maximum number of hops.

Part IV – NSLookup Command

15. Execute the nslookup command to resolve a domain name into an IP address.
16. Perform reverse DNS lookup using an IP address.
17. Query different DNS record types such as **A** and **AAAA** records.
18. Verify DNS server functionality.

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Part V – Netstat Command

19. Execute the netstat command to display active network connections.
20. Use netstat -a to display all active and listening ports.
21. Execute netstat -n, netstat -o, netstat -e, netstat -r, netstat -s, and netstat -ano.
22. Observe protocol statistics, routing information, Ethernet statistics, and Process IDs (PID).

Part VI – ARP Command

23. Execute the arp -a command to display the ARP cache.
24. Observe the IP-to-MAC address mappings.
25. Delete ARP entries using arp -d.
26. Create a static ARP entry using arp -s.

Part VII – Hostname Command

27. Execute the hostname command.
28. Observe the current computer name used on the network.

Part VIII – PathPing Command

29. Execute the pathping command for a remote host.
30. Observe the complete route, packet loss percentage, and latency at each router.
31. Identify any router causing delays or packet loss.

Part IX – Route Command

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32. Execute the route print command to display the routing table.
33. Add a static route using the route add command.
34. Delete the static route using the route delete command.
35. Observe the changes in the routing table.

Part X – Net Use Command

36. Execute the net use command to display currently connected network resources.
37. Map a shared folder to a drive letter using the net use command.
38. Disconnect the mapped drive using net use /delete.
39. Verify that the network resource has been removed successfully.

DEMONSTRATION/OUTPUT:

Part-1: Ping command

```
> ping google.com
PING google.com (142.251.106.139) 56(84) bytes of data:
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=1 ttl=109 time=357 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=2 ttl=109 time=277 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=3 ttl=109 time=239 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=4 ttl=109 time=428 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=5 ttl=109 time=77.1 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=6 ttl=109 time=112 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=7 ttl=109 time=290 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=8 ttl=109 time=71.6 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=10 ttl=109 time=203 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=11 ttl=109 time=226 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=12 ttl=109 time=249 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=13 ttl=109 time=375 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=14 ttl=109 time=400 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=15 ttl=109 time=214 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=16 ttl=109 time=343 ms
```

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```
~ via d v3.14.6
> ping 127.25.189.232
PING 127.25.189.232 (127.25.189.232) 56(84) bytes of data.
64 bytes from 127.25.189.232: icmp_seq=1 ttl=64 time=0.052 ms
64 bytes from 127.25.189.232: icmp_seq=2 ttl=64 time=0.037 ms
64 bytes from 127.25.189.232: icmp_seq=3 ttl=64 time=0.036 ms
64 bytes from 127.25.189.232: icmp_seq=4 ttl=64 time=0.037 ms
64 bytes from 127.25.189.232: icmp_seq=5 ttl=64 time=0.044 ms
64 bytes from 127.25.189.232: icmp_seq=6 ttl=64 time=0.047 ms
64 bytes from 127.25.189.232: icmp_seq=7 ttl=64 time=0.036 ms
64 bytes from 127.25.189.232: icmp_seq=8 ttl=64 time=0.036 ms
64 bytes from 127.25.189.232: icmp_seq=9 ttl=64 time=0.041 ms
64 bytes from 127.25.189.232: icmp_seq=10 ttl=64 time=0.041 ms
64 bytes from 127.25.189.232: icmp_seq=11 ttl=64 time=0.371 ms
64 bytes from 127.25.189.232: icmp_seq=12 ttl=64 time=0.043 ms
64 bytes from 127.25.189.232: icmp_seq=13 ttl=64 time=0.040 ms
64 bytes from 127.25.189.232: icmp_seq=14 ttl=64 time=0.032 ms
64 bytes from 127.25.189.232: icmp_seq=15 ttl=64 time=0.037 ms
64 bytes from 127.25.189.232: icmp_seq=16 ttl=64 time=0.038 ms
64 bytes from 127.25.189.232: icmp_seq=17 ttl=64 time=0.036 ms
```

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```

> ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp3s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether d8:43:ae:07:5c:8f brd ff:ff:ff:ff:ff:ff
    altname enx843ae075c8f
3: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 8e:06:4d:a4:f5:bc brd ff:ff:ff:ff:ff:ff permaddr e0:c2:64:ac:36:86
    altname wlp0s20f3
    altname wlxe0c264ac3686
    inet 172.25.189.232/20 brd 172.25.191.255 scope global dynamic noprefixroute wlo1
        valid_lft 2834sec preferred_lft 2834sec
    inet6 fe80::5ee8:7037:208a:74a5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

```

> ip maddress
1: lo
    inet 224.0.0.251
    inet 224.0.0.1
    inet6 ff02::fb
    inet6 ff02::1
    inet6 ff01::1
2: enp3s0
    link 33:33:00:00:00:01
    link 01:00:5e:00:00:01
    inet 224.0.0.1
    inet6 ff02::1
    inet6 ff01::1
3: wlo1
    link 01:00:5e:00:00:01 users 2
    link 01:00:5e:00:00:fb users 2
    link 33:33:00:00:00:01 users 2
    link 33:33:ff:8a:74:a5 users 2
    link 33:33:00:00:00:fb users 2

```


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Part-3: traceroute Command

```
~ via d v3.14.6
> traceroute www.google.com
traceroute to www.google.com (142.251.150.119), 30 hops max, 60 byte packets
 1 _gateway (172.25.176.1)  7.684 ms  11.318 ms  11.308 ms
 2 192.168.50.11 (192.168.50.11)  7.693 ms  7.683 ms  7.674 ms
 3 192.168.50.1 (192.168.50.1)  5.826 ms  5.817 ms  7.637 ms
 4 172.16.2.1 (172.16.2.1)  5.801 ms  5.790 ms  5.782 ms
|
```

```
^C
~ via d v3.14.6 took 26s
> traceroute -d 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
setsockopt SO_DEBUG: Permission denied

~ via d v3.14.6
> |
```

```
> traceroute --help
Usage:
  traceroute [ -4dFItnreAUDV ] [ -f first_ttl ] [ -g gate,... ] [ -i device ] [ -m max_ttl ] [ -N
[ -q nqueries ] [ -s src_addr ] [ -z sendwait ] [ --fwmark=num ] host [ packetlen ]
Options:
-4                                Use IPv4
-6                                Use IPv6
-d --debug                        Enable socket level debugging
-F --dont-fragment                Do not fragment packets
-f first_ttl --first=first_ttl    Start from the first_ttl hop (instead from 1)
-g gate,... --gateway=gate,...    Route packets through the specified gateway
                                  (maximum 8 for IPv4 and 127 for IPv6)
-I --icmp                        Use ICMP ECHO for tracerouting
-T --tcp                          Use TCP SYN for tracerouting (default port is 80)
-i device --interface=device      Specify a network interface to operate with
-m max_ttl --max-hops=max_ttl     Set the max number of hops (max TTL to be
                                  reached). Default is 30
-N squeries --sim-queries=squeries Set the number of probes to be tried
                                  simultaneously (default is 16)
-n                                Do not resolve IP addresses to their domain names
-p port --port=port              Set the destination port to use. It is either
                                  initial udp port value for "default" method
                                  (incremented by each probe, default is 33434), or
                                  initial seq for "icmp" (incremented as well,
                                  default from 1), or some constant destination
                                  port for other methods (with default of 80 for
                                  "tcp", 53 for "udp", etc.)
-t tos --tos=tos                 Set the TOS (IPv4 type of service) or TC (IPv6
```

Part-4: NSLookup Command

```
~ via d v3.14.6
> dig google.com

; <<>> DiG 9.18.50 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39248
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                198     IN      A      142.250.118.138
google.com.                198     IN      A      142.250.118.100
google.com.                198     IN      A      142.250.118.101
google.com.                198     IN      A      142.250.118.113
google.com.                198     IN      A      142.250.118.139
google.com.                198     IN      A      142.250.118.102

;; Query time: 9 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Jul 22 10:45:08 IST 2026
;; MSG SIZE rcvd: 135

~ via d v3.14.6
>
```

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Part-5: Netstat Command

```

> netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address          State
tcp        0      0 fedora:49456            maa05s23-in-f10.1:https TIME_WAIT
tcp        0      0 fedora:49478            maa05s23-in-f10.1:https ESTABLISHED
tcp        0      0 fedora:39166            ec2-18-97-36-75.c:https ESTABLISHED
tcp        0      0 fedora:37922            141.207.186.35.bc:https TIME_WAIT
tcp        0      0 fedora:35074            instagram-p3-shv-:https ESTABLISHED
tcp        0      0 fedora:58154            lb-140-82-114-26-:https ESTABLISHED
tcp        0      0 fedora:60170            lcdelw-in-f94.1e1:https ESTABLISHED
tcp        0      0 fedora:58586            104.20.28.164:https     ESTABLISHED
tcp        0      0 fedora:33612            hz16gb0.beaondb.:https  TIME_WAIT
tcp        0      0 fedora:33678            hz16gb0.beaondb.:https  TIME_WAIT
tcp        0      0 fedora:53418            nw-in-f139.1e100.:https ESTABLISHED
tcp        0      0 fedora:45994            172.67.206.89:https     ESTABLISHED
tcp        0      0 fedora:48882            whatsapp-cdn-shv-:https ESTABLISHED
tcp        0      0 fedora:36822            142.251.150.119:https  ESTABLISHED
tcp        0      0 fedora:52200            lcdels-in-f100.1e:https TIME_WAIT
tcp        0      0 fedora:52218            lcdels-in-f100.1e:https TIME_WAIT
tcp        0      0 fedora:35418            nu-in-f139.1e100.:https TIME_WAIT
tcp        0      0 fedora:35452            nu-in-f139.1e100.:https ESTABLISHED
tcp        0      0 fedora:46368            kix05s01-in-f10.1:https TIME_WAIT
tcp        0      0 fedora:46374            kix05s01-in-f10.1:https ESTABLISHED
tcp        0      0 fedora:48386            tzdelb-bg-in-f3.1:https TIME_WAIT
tcp        0      0 fedora:50804            lcboma-bi-in-f10.:https ESTABLISHED
tcp        0      0 fedora:34738            edge-dgw-shv-02-b:https ESTABLISHED
tcp        0      0 fedora:34760            edge-dgw-shv-02-b:https ESTABLISHED
tcp        0      0 fedora:34770            edge-dgw-shv-02-b:https ESTABLISHED
tcp        0      0 fedora:51784            93.243.107.34.bc.:https ESTABLISHED
tcp        0      0 fedora:40584            edge-dgw-shv-02-b:https ESTABLISHED
tcp        0      0 fedora:39084            tzdelb-au-in-f14.:https TIME_WAIT
tcp        0      0 fedora:51208            nw-in-f113.1e100.:https TIME_WAIT
tcp        0      0 fedora:60688            lcdels-in-f84.1e1:https ESTABLISHED

```

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Part-6: ARP Command

```
~ via d v3.14.6 took 7s  
> arp -a  
_gateway (172.25.176.1) at cc:5a:53:d3:d5:68 [ether] on wlo1  
  
~ via d v3.14.6
```

Part-7: Hostname

```
~ via d v3.14.6  
> hostname  
fedora  
  
~ via d v3.14.6
```

Part-8: Pathping Command

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My traceroute [v0.96] 2026-07-22T10:47:35+0530

fedora (172.25.189.232) -> google.com (142.250.118.113)

Keys: Help Display mode Restart statistics Order of fields quit

Host	Packets		Pings					
	Loss%	Snt	Last	Avg	Best	Wrst	StDev	
1. _gateway	0.0%	2	43.5	23.5	3.4	43.5	28.3	
2. 192.168.50.11	0.0%	2	3.1	4.9	3.1	6.8	2.6	
3. 192.168.50.1	0.0%	2	7.2	4.8	2.5	7.2	3.3	
4. 172.16.2.1	0.0%	2	2.8	3.0	2.8	3.2	0.2	
5. (waiting for reply)								
6. (waiting for reply)								
7. (waiting for reply)								
8. (waiting for reply)								
9. (waiting for reply)								
10. 72.14.196.97	0.0%	2	136.0	112.1	88.2	136.0	33.8	
11. 142.251.77.185	0.0%	2	131.4	128.2	125.1	131.4	4.4	
12. 192.178.83.224	0.0%	2	125.0	114.4	103.7	125.0	15.0	
13. 142.251.234.11	0.0%	2	80.5	88.6	80.5	96.7	11.5	
14. 216.239.48.78	0.0%	2	72.7	75.4	72.7	78.0	3.7	
15. (waiting for reply)								
16. (waiting for reply)								
17. (waiting for reply)								
18. (waiting for reply)								
19. (waiting for reply)								
20. (waiting for reply)								
21. nu-in-f113.1e100.net	0.0%	1	60.8	60.8	60.8	60.8	0.0	

Part-9: Route Command

```
~ via d v3.14.6
> mtr google.com

~ via d v3.14.6 took 28s
> route print
Usage: route [-nNvee] [-FC] [<AF>]          List kernel routing tables
        route [-v] [-FC] {add|del|flush} ... Modify routing table for AF.

        route {-h|--help} [<AF>]             Detailed usage syntax for specified AF.
        route {-V|--version}                 Display version/author and exit.

        -v, --verbose                         be verbose
        -n, --numeric                         don't resolve names
        -e, --extend                           display other/more information
        -F, --fib                             display Forwarding Information Base (default)
        -C, --cache                           display routing cache instead of FIB

<AF>=Use -4, -6, '-A <af>' or '--<af>'; default: inet
List of possible address families (which support routing):
  inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
  netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
  x25 (CCITT X.25)
```

LEARNING OUTCOMES:

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After completing this experiment:

- I understood the purpose and working of essential Windows networking commands such as Ping, IPConfig, Tracert, NSLookup, Netstat, ARP, Hostname, PathPing, Route, and Net Use, and learned how they are used for network configuration, monitoring, and troubleshooting.
- I gained practical knowledge of network connectivity testing by using the Ping command and its various options to verify communication between devices, measure network latency, identify packet loss, and diagnose basic network connectivity issues.
- I learned to view and manage TCP/IP network configuration using IPConfig, trace packet routes using Tracert, resolve domain names using NSLookup, and analyze routing paths and packet loss using PathPing, enabling effective diagnosis of network communication problems.
- I developed the ability to monitor and analyze network activity using Netstat, understand IP-to-MAC address mapping through ARP, identify the system using the Hostname command, and manage routing information and shared network resources using the Route and Net Use commands.
- I enhanced my practical skills in network administration and troubleshooting by executing command-line networking utilities in Windows, interpreting their outputs, identifying common network issues, and applying appropriate commands to verify connectivity, resolve configuration problems, and manage network resources efficiently.